Linshan Ye

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Education

Tsinghua University, Beijing, China *2023 – 2027 (expected)*
Bachelor of Science in Biological Science
Minor in Statistics and Data Science

- **GPA:** 3.9/4.0 **Ranking:**3/35
- **Relevant Coursework (Biological Science):** Data Science in Microbiology (A+), Public health (P), Chemical Biology (A-), Biochemistry(1) (A), Biochemistry(2) (A-), Cell Biology (A), Molecular Biology (A-), Genetics (A), Developmental Biology (A-), Basic Practical Biochemistry (A), Fundamental Microbiology Experiments (A-), Integrated Practice of General Biology in Field (A-), Laboratory of Molecular Biology (A)
- **Relevant Coursework (Statistics & Data Science):** Convex Optimization, Random Mathematics and Statistics, Design and Analysis of Experiments, Monte Carlo Methods(in progress, Fall 2026), Statistical Computing and Software (P), Programming for Data Science (A-), Multivariate Statistical Analysis (A-), Statistical Machine Learning (A), Statistical Inference (P), Linear Regression Analysis (A), Elementary Probability Theory (A-), Linear Algebra (A-), Calculus B1 (B+), Calculus B2 (A)

Westlake University, Hangzhou, China *2025*
"Frontiers of Life Sciences" International Undergraduate Summer School

Research Interests

Applied data science and machine learning for **computational biology, machine learning, AI for science**.

Research Experience

Construction of a SIRT2 Target Prediction Model Based on Deep Learning Classifiers and Protenix Fine-Tuning
07 2026 – 09 2026
Advisor: Prof. Hening Lin *University of Chicago, Chicago*

- Reframed SIRT2 substrate identification as a binary classification problem using 6,802 data entries from an experimentally annotated SIRT2 peptide microarray dataset (doi: 10.1038/ncomms3327), to resolve high false-positive rates in existing structure prediction models such as Protenix.
- Trained a multilayer perceptron (MLP) classifier to learn SIRT2 substrate sequence preferences, and integrated it downstream of Protenix for end-to-end joint fine-tuning.
- Achieved an area under curve (AUC) of 0.73 with the standalone MLP classifier, and improved AUC to 0.783 after joint fine-tuning with Protenix.
- Applied the fine-tuned model to a human proteome-wide screen of lysine-containing peptides to prioritize candidate substrates for experimental validation. Six top candidates from plasmids available in the lab were selected for subsequent functional testing, and two were previously unreported SIRT2 substrates.

Research on Privacy-Preserving Machine Unlearning Algorithms Based on Synthetic Data Augmentation
03 2026 – Present
Advisor: Prof. Sai Li *Tsinghua University, Beijing*

- Developed a knowledge-graph-guided data augmentation pipeline that could extract target facts from source corpora and generate diverse, fact-preserving rewrites for large language model (LLM) unlearning.
- Applied negative preference optimization (NPO) and gradient difference (GDiff) unlearning to original and rewritten corpora, then retrained on AGNews and SST-2 datasets to assess robustness to benign relearning using WMDP-Bio benchmark's forget accuracy.
- Reduced post-relearning WMDP-Bio forget accuracy from approximately 58% to 28–48% with NPO and from 62% to 58% with GDiff, compared with unlearning on original corpora.
- Ongoing work extends the evaluation to MUSE-Books and MUSE-News benchmark, using both NPO and GDiff.

Development of a Bacteriophage-Editing Toolkit in Gut Bacteroidales
06 2024 – Present
Advisor: Prof. Guanxiang Liang *Tsinghua University, Beijing*

- Co-developed a CRISPR–Cas12a-based phage engineering system for Bacteroidales, a prevalent gut bacterial clade lacking reliable genetic manipulation tools.
- Designed and built Bacteroidales-compatible plasmids expressing Cas12a and phage-targeting crRNAs, which could enable targeted genomic cleavage and template-directed sequence replacement upon phage infection.
- Achieved robust, target-specific genome cleavage in *P. distasonis*, establishing the first functional CRISPR phage

editing toolkit for Bacteroidales.
- Ongoing work extends the toolkit to additional Bacteroidales species (*B. kribbi* and *B. thetaiotaomicron*) for gut microbiome manipulation studies.

Manuscript in Preparation

Manuscript in preparation: Improving Robustness To Benign Relearning Via Knowledge-Dense Unlearning (with Prof. Sai Li).

Technical Skills

Programming:	Python (NumPy, pandas, scikit-learn, PyTorch), R, Git, Linux/HPC (Slurm), LaTeX, batch queries
Data Science:	MLP, model fine-tuning, EM algorithm, linear regression, unsupervised clustering, knowledge-graph construction, LLM fine-tuning
Drylab Biology:	protein structure prediction (Protenix/AlphaFold), genome annotation, genome assembly
Wetlab Biology:	PCR, Western blot, cell/bacteria/phage/fungi culture, electrophoresis, DNA/Protein extraction

Honors and Awards

International Genetically Engineered Machine Competition (iGEM) 2025 Gold Medal, nominated for Best Sustainable Development Impact	2025
Xuetang Talent Development Program Scholarship, Tsinghua University	2025
Xuetang Talent Development Program Scholarship, Tsinghua University	2024
Comprehensive Excellence Scholarship, Tsinghua University	2024
Academic Excellence Scholarship, Zhili College, Tsinghua University	2024

Academic and Social Practice

- **Third Prize (Final)**, 20th "Novonesis Cup" Seven Capital Universities Life Sciences Culture Festival *2025*
- **Student Co-Founder**, Tsinghua University Synthetic Biology Student Science and Technology Interest Team *2025*
- **Oral Presentation**, Capital Universities 2024 Annual Conference on Biological Field Practice Exchange *2024*
- **Student Leader**, Department of Social Practice, University Youth League Committee *2025-Present*
- **Student Team Leader**, International Social Practice Team to Jakarta and Bali, Indonesia *2025*
- **Student Team Vice Leader**, Domestic Social Practice Team to Tianjin, China *2024*

Standardized Test Scores

GRE General Test (July 14, 2023): Verbal Reasoning 160/170; Quantitative Reasoning 169/170; Analytical Writing 4.0/6.0.

TOEFL iBT (September 13, 2026): Overall 6.0/6.0; Reading 6.0, Listening 6.0, Writing 5.0, Speaking 6.0.